Homework 2

MATH 518 - Fall 2025

Due on Friday, October 10, 11:59pm on Crowdmark

Total of points = 50 points

Please submit each question separately on Crowdmark. Unless stated otherwise, k is an algebraically closed field of characteristic zero.

1. Let $I = (x^2 - y^3, y^2 - z^3) \subset k[x, y, z]$ and $V = V(I)$. Define the homomorphism of k-algebras $\alpha\colon k[x, y, z] \longrightarrow k[t]$ by $\alpha(x) = t^9$, $\alpha(y) = t^6$ and $\alpha(z) = t^4$.
   (a) (5 points) Show that the kernel of α is I. Deduce that V is irreducible.
   (b) (5 points) The previous map induces a homomorphism $\overline{\alpha}\colon k[V] \longrightarrow k[t]$ of k-algebras. What is the morphism $\varphi\colon k \longrightarrow V$ of affine varieties that corresponds to $\overline{\alpha}$ *i.e.* such that $\varphi^* = \overline{\alpha}$? Show that φ is one-to-one and onto, but not an isomorphism.

Solution:

(a) Any $f \in k[x, y, z]$ is congruent modulo I to a polynomial

$$f(x, y, z) = a(z) + b(z)x + c(z)y + d(z)xy, \tag{1}$$

since the higher powers of x and y can be written as a power of z. If f is in the kernel of α, then for all $t \in k$

$$f(t^9, t^6, t^4) = a(t^4) + b(t^4)t^9 + c(t^4)t^6 + d(t^4)t^{15} = 0. \tag{2}$$

But the degrees appearing in each of these four polynomials are all distinct: the first one only has degrees congruent to 0 mod 4, the second one only has degrees congruent to 9 mod 4, etc. So there is no possible cancellation, and we deduce that $a = b = c = d = 0 \in k[z]$. This shows that $\ker(\alpha) = I$. Since the kernel is I, the ring $k[V]$ is a subring of $k[t]$. The latter being an integral domain, the ring $k[V]$ is also an integral domain and V is irreducible.

(b) The map φ is

$$\varphi\colon k \longrightarrow V \qquad t \longmapsto (t^9, t^6, t^4). \tag{3}$$

Let S be the image of φ. We want to show that $S = V$. It is clear that $S \subset V$, by plugging in (t^9, t^6, t^4) in the polynomial equations. Conversely, if $(x, y, z) \in V$ then $x^2 = y^3$ and $y^2 = z^3$. Let us write $z = t^4$ for some $t \in k$ (this is possible since k is algebraically closed, and there are four possible choices). Then $y^2 = t^{12}$ implies that $y = \pm t^6$. If $y = t^6$ we are done. If $y = -t^6$ then we can replace t by $u = it$ and we have $z = u^4$ and $y = u^6$. Similarly, we deduce from $x^2 = y^3 = u^{18}$ that $x = \pm u^9$. If $x = u^9$ we are done since we showed that $(x, y, z) = (u^9, u^6, u^4)$. If $x = -u^9$ then we replace u by $-u$ to show that it is of the desired form. We have shown that $V = S$, so in particular that φ is onto. To see that it is one-to-one, assume

$$u^4 = t^4, \quad u^6 = t^6, \quad u^9 = t^9. \tag{4}$$

for two distinct u and t (in particular both are nonzero). Then $(u/t)^4 = 1$ and $1 = (u/t)^9 = (u/t)((u/t)^4)^2 = u/t$ shows that the map is also one-to-one. Finally, the map is not an isomorphism of varieties since $\overline{\alpha}$ is not an isomorphism of k-algebras: it is not surjective. Indeed, the image in $k[t]$ only contains polynomials of the form $f(t^9, t^6, t^4)$. In particular, we can't get any polynomials of degrees $1, 2$ or 3 in $k[t]$.

2. (10 points) For each of the following functions, determine at which points of the affine variety V they are regular.
   (a) (5 points) The function $f = \frac{1-\overline{y}}{\overline{x}}$ on $V = V(x^2 + y^2 - 1) \subset k^2$
   (b) (5 points) The function $f = \frac{\overline{x}}{\overline{y}}$ on $V = V(xw - yz) \subset k^4$.

Solution:

(a) The function is regular if $\overline{x} \neq 0$. If $\overline{x} = 0$, then $\overline{y} = \pm 1$. At the point $(0, 1)$ the function is regular, since

$$\frac{1 - \overline{y}}{\overline{x}} = \frac{\overline{x}}{1 + \overline{y}}. \tag{5}$$

Let's show that the function is not defined at the point $(0, -1)$. Suppose we had polynomials $f, g \in k[x, y]$ such that

$$\frac{\overline{x}}{1 + \overline{y}} = \frac{\overline{f}}{\overline{g}}$$

and $g(0, -1) \neq 0$. Then this would mean that

$$xg(x, y) - f(x, y)(1 + y) = P(x, y)(x^2 + y^2 - 1) \tag{6}$$

for some polynomial $P \in k[x, y]$. Modulo $I(V)$ (since $y^2 = 1 - x^2$), we can assume that the representatives f, g are of the form

$$\begin{aligned} f(x,y) &= f_0(y) + x f_1(y) \\ g(x,y) &= g_0(y) + x g_1(y) \end{aligned}$$

where $f_0, f_1, g_0, g_1 \in k[y]$. The condition $g(0, -1) \neq 0$ implies that

$$g_0(-1) \neq 0. \tag{7}$$

Pluging in the previous equation (6) gives

$$\begin{aligned} P(x,y)(x^2+y^2-1) &= [g_0(y) + x g_1(y)]\, x - (1+y)\,[f_0(y) + x f_1(y)] \\ &= g_1(y)x^2 + x\,[g_0(y) - (1+y)f_1(y)] - (1+y)f_0(y). \end{aligned}$$

By comparing both sides we see that $P(x, y) = g_1(y)$, and so there is no term of degree 1 in x on the LHS. Thus

$$g_0(y) = (1+y)f_1(y),$$

which contradicts the assumption $g_0(-1) \neq 0$.

In conclusion, the only pole of f is at $(0, -1)$.

(b) The function is defined when $\overline{y} \neq 0$. If $\overline{y} = 0$, then $\overline{z} = 0$ or $\overline{w} = 0$. By the relation $\frac{\overline{x}}{\overline{y}} = \frac{\overline{z}}{\overline{w}}$, we see that the function is defined if $\overline{w} \neq 0$. Thus, the set of poles is contained in $S = \{(\overline{x}, 0, \overline{z}, 0) \in k^4\}$. We claim that these are exactly the poles. Suppose there is a $p \in S$ such that

$$f = \frac{\overline{x}}{\overline{y}} = \frac{\overline{g}}{\overline{h}} \tag{8}$$

for some polynomials $g, h \in k[x, y, w, z]$, and such that $h(x, 0, z, 0) \neq 0$. This is equivalent to

$$xh - yg \in (xw - yz). \tag{9}$$

so there is a polynomial $P \in k[x, y, w, z]$ such that $xh - yg = P(xw - yz)$. This can be rewritten

$$y(Pz - g) = x(Pw - h). \tag{10}$$

Since x is irreducible, it implies that y divides $Pw - h$, so that

$$h(x,y,z,w) = P(x,y,z,w)w - Q(x,y,z,w)y \tag{11}$$

for some $Q \in k[x, y, z, w]$. Evaluating at $(x, 0, z, 0)$ implies that $h(x, 0, z, 0) = 0$, which is a contradiction to the previous assumption.

3. (a) (5 points) Find the singular points of the following curves

   1. $x^4 + y^4 - x^2y^2$
   2. $x^3 + y^3 - 3x^2 - 3y^2 + 3xy + 1$

   Hint: simplify the polynomial

   (b) (5 points) Show that an irreducible plane curve only has finitely many singular points.

Solution:

(a) 1. Let $f = x^4 + y^4 - x^2y^2$. A point $p \in V(f)$ is a multiple point if both derivatives

$$f_x = 4x^3 - 2xy^2 = 2x(2x^2 - y^2)$$
$$f_y = 4y^3 - 2y^2x = 2y(2y^2 - x^2)$$

vanish. From the first equation we find $x = 0$ or $y^2 = 2x^2$. If $x = 0$, then the second equation shows that $y = 0$ and so $(0,0) \in V(f)$ is a singular point. On the other hand, if $y^2 = 2x^2$ then we also find from the second equation that $x = y = 0$. Thus $(0,0)$ is the only singular point.

2. Note that $f = x^3+y^3-3x^2-3y^2+3xy+1 = (x-1)^3+3(x-1)(y-1)+(y-1)^3$. The derivatives are

$$f_x = 3(x-1)^2 + 3(y-1)$$
$$f_y = 3(y-1)^2 + 3(x-1).$$

If $f_x = 0$ then $(y-1) = -(x-1)^2$, and plugging in $f_y = 0$ gives

$$0 = (x-1)^4 + (x-1) = (x-1)((x-1)^3 + 1).$$

So either $x = 1$ or $x = 0$. From $y = 1 - (x-1)^2$ we find the two possible points $(0,0)$ and $(1,1)$. However, the point $(0,0)$ is not in $V(f)$, so $(1,1)$ is the only singular point of f.

(b) Let f be irreducible, and f_x, f_y the derivatives in x, y. The set of singular points is exactly $V(f, f_x, f_y) = V(f, f_x) \cap V(f, f_y)$. We have seen that if two polynomials $f, g \in k[x,y]$ have no common factors, then $V(f,g)$ is finite. Since f is irreducible and the degrees of f_x, f_y are lower than f, the sets $V(f, f_x)$ and $V(f, f_y)$ are both finite.

4. (10 points) Let $f\colon k^3 \longrightarrow k^3$ be the regular map $f(x,y,z) = (x, xy, xyz)$. Find the image in k^3. Is it Zariski closed? open? dense?

Hint: the intersection of two closed sets is closed. If a polynomial $g \in k[x]$ has infinitely many roots then $g = 0$.

Solution: If $x = 0$, the image is $(0, 0, 0)$. If $x \neq 0$ and $y = 0$, the image is $(0, 0, 0)$. Thus, the image of this map is the set

$$\begin{aligned} S &= \left\{(x, y, z) \in k^3 \mid x \neq 0 \text{ and } y \neq 0\right\} \cup V(y, z) \\ &= k^* \times k^* \times k \cup V(y, z) \\ &= \left(k^3 \smallsetminus V(xy)\right) \cup V(y, z). \end{aligned}$$

It consists of the x-axis $V(y, z)$ and all points in $k^3 \smallsetminus V(y, z)$ that are not on the yz-plane nor the xz-plane.

The set S is not closed. If S was closed, then the intersection with any closed set A would be closed in A. We see that if we intersect S with the line $A = V(y-1, z)$, then

$$S \cap V(y-1, z) = \left\{(x, 1, 0) \in k^3 \ x \neq 0\right\}. \tag{12}$$

However this is a line minus the origin, so it is not closed (its complement is closed). It follows that S is not closed.

The set S is not open. If it was open, then its complement $k^3 \smallsetminus S = V(xy) \smallsetminus V(y, z)$ would be closed. Note that this set is the union of the xz-plane and the yz-plane, minus the x-axis. But the intersection with the y-axis $V(x, z)$ is $V(x, z) \smallsetminus \{(0, 0, 0)\}$ (the y-axis minus the origin), which is open in $V(x, z)$.

Finally, let us show that the set is dense, that is that the closure is $\overline{S} = k^3$. Since $X = k^3 \smallsetminus V(xy) \subset S$, it is enough to show that

$$\overline{X} = k^3.$$

Recall that $\overline{X} = V(I(X))$. We are done if show that $I(X) = 0$. So suppose that that $f \in I(X)$ is a nonzero polynomial. Thus we have $f(x, y, z) = 0$ for all $z \in k$ and $y \in k^*$. If we fix some $y = b$ and $c = z$, then

$$f(x, b, c) = f_0(b, c) + f_1(b, c)x + \cdots + f_n(b, c)x^n \in [x] \tag{13}$$

is a polynomial with infinitely many zeros. This implies that the polynomials $f_i(y, z) = 0$ are zero for all $y \in k^*$ and $z \in k$. Repeating the same argument for the f_i shows that $f = 0$, which is a contradiction.

5. (10 points) Let $\varphi \colon V \longrightarrow W$ be a morphism of varieties, and $\varphi^* \colon k[W] \longmapsto k[V]$ the induced k-algebra morphism. Let $p \in V$ and $q \in W$ two points such that $\varphi(p) = q$. Show that φ^* extends uniquely to a ring morphism

$$\varphi^* \colon \mathcal{O}_q(W) \longrightarrow \mathcal{O}_p(V),$$

and that $\varphi^*(\mathfrak{m}_q(W)) \subset \mathfrak{m}_p(V)$.
Remark: In general, φ^* does not extend to morphism $k(W) \longrightarrow k(V)$.

Solution: Let $f \in \mathcal{O}_q(W)$. Then we can write $f = \frac{\overline{a}}{\overline{b}}$ for some $\overline{a}, \overline{b} \in k[W]$ such that $\overline{b}(q) \neq 0$. We define

$$\varphi^*(f) = \frac{\varphi^*(\overline{a})}{\varphi^*(\overline{b})}. \tag{14}$$

Note that $\varphi^*(\overline{b})(p) = \overline{b}(\varphi(p)) = \overline{b}(q) \neq 0$ so that $\varphi^*(f) \in \mathcal{O}_p(V)$. (In particular it implies that $\varphi^*(\overline{b}) \neq 0$, which is not necessarily the case if one tries to extend this map to $k(W) \to k(V)$).

We have to show that the map does not depend on the choice of the representative. Suppose that $f = \frac{\overline{a}}{\overline{b}} = \frac{\overline{c}}{\overline{d}}$ with $\overline{d}(q) \neq 0$. Since φ^* is a morphism of k-algebras, we have

$$0 = \varphi^*(0) = \varphi^*(\overline{a}\overline{d} - \overline{b}\overline{c}) = \varphi^*(\overline{a})\varphi^*(\overline{d}) - \varphi^*(\overline{b})\varphi^*(\overline{c}),$$

and so

$$\frac{\varphi^*(\overline{a})}{\varphi^*(\overline{b})} = \frac{\varphi^*(\overline{c})}{\varphi^*(\overline{d})}.$$

Finally, if $f \in \mathfrak{m}_q(W)$ then $\overline{a}(q) = 0$ and so $\varphi^*(\overline{a})(p) = \overline{a}(\varphi(p)) = \overline{a}(q) = 0$. Thus $\varphi^*(\mathfrak{m}_q(W)) \subset \mathfrak{m}_p(W)$.